Isaac Watts

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Education

Virginia State University (GPA: 3.9) — *May 2025*
Bachelor of Science: Computer Science
Bachelor of Science: Mathematics

Technical Skills

Programming Languages: Python, C, C++, Java, C#, Kotlin, MATLAB, JavaScript, TypeScript, LaTeX

Databases: Relational databases (MySQL, PostgreSQL, and SQLite) and Nonrelational databases (MongoDB)

Data Processing & BI Tools: Tableau, Power BI, Excel, Pandas, NumPy, Matplotlib, PyTorch, SciPy

Version Control & Containerization: Git, GitLab, Docker

Operating Systems: Windows 11 and Ubuntu 22.04 (Linux)

Experience

Freelance App Developer — *Petersburg, VA*
PNIRD — *January 2025 – present*

- Integrated OpenAI embeddings, PostgreSQL, and cosine similarity to reduce query response time from seconds to milliseconds.
- Developed a cross-platform medical app in React Native (TypeScript/JavaScript) with a Python (Pandas) ETL pipeline to improve healthcare data accessibility.
- Designed a CI/CD pipeline with GitHub Actions to automate builds, testing, and deployment for future App Store releases.

Robotics Student Researcher — *Adelphi, MD*
Army Research Lab — *May 2023 – August 2023*
May 2024 – August 2024

- Trained reinforcement learning agents in Unity (C#, ML-Agents, PyTorch, SciPy) to restore radio communications, containerized with Docker for reproducibility, and version control using Git with GitLab.
- Built an image processing utility in C++ (ROSCPP) to convert/compress 8-bit → 16-bit images for efficient network transfer.
- Automated RVIZ visualization file generation with Python (ROSPY), eliminating manual setup and boosting research productivity.
- Maintained and optimized ROS1 Noetic on Ubuntu to streamline multi-robot experiment setup and ensure reproducible environments.

Projects

Data Visualizations from CDC dataset — *Tableau Dashboard*

- Created a multi-page interactive Tableau dashboard analyzing chronic disease trends by state, gender, and demographics, highlighting health disparities and key insights.

Inverse Problem Modeling and Machine Learning — *github.com/IwattsX/med_bodies*

- Simulated voltage readings in MATLAB, trained a neural net using PyTorch with ELU activations functions and the Adam optimizer to localize inclusions from voltage patterns.

Robotic Arm Simulation using Isaac Sim — *github.com/stars/IwattsX/lists/robotsim*

- Visualized Oculus Quest hand data in Python/Matplotlib to simulate in Isaac Sim.